MID EVALUATION

C3: Committed Confidence Calibration

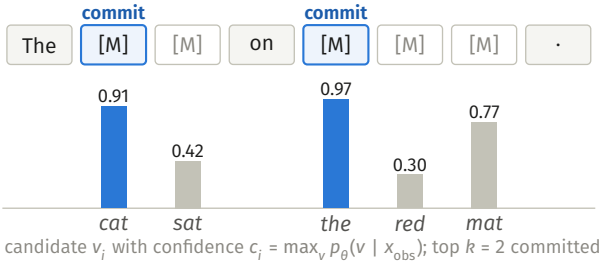
Selection bias in confidence-based token commitment in diffusion language models

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Diffusion LMs commit the tokens they are most confident about

One forward pass scores every masked position; the decoder keeps the top-k



- LLaDA-8B and Dream-7B keep the most confident positions (MaskGIT-style); Fast-dLLM puts one global threshold on the same number.
- A committed token won: its confidence was the largest of m noisy estimates. That is the setting of the winner's curse (Efron 2011; Andrews et al. 2024).

Question. Once a token has won, is its confidence still an honest probability of being right? Threshold papers report speed and quality, never calibration.

Two separate ways parallel decoding goes wrong

For a set S committed in one step, given the observed context x_{obs}

$$\mathrm{KL}\Big(p^* \,\Big\|\, \prod_{i\in S} p_{\theta,i}\Big) = \underbrace{\mathrm{TC}(x_S \mid x_{\text{obs}})}_{\text{(A) dependency}} + \underbrace{\sum_{i\in S} \mathrm{KL}\big(p_i^* \,\|\, p_{\theta,i}\big)}_{\text{(B) estimation}}$$

- **(A)** Marginals, never the joint: tokens can each be plausible yet jointly nonsensical.
- **(B)** The marginals themselves are wrong. The dependency literature treats (B) as fixed.
- Selecting on a noisy estimate is one way (B) could be large exactly where a decoder commits.

Work	Calibrated under selection?	Separates (A) and (B)?
Fast-dLLM; OSDT	no	no
KLASS; SWD; LESS	no	no
PC-Sampler	positional only	no
DAPD; DEMASK; PUNT	no	(A) only
ParallelBench	no	(A) only
Tang et al. (2026)	no	identifies
This work	**yes**	**both**

Fixes to our proposal

Proposition 1 · Winning adds nothing given confidence and context

If the pairs (ℓ_j, a_j) are independent across $j \in M$ given s, and the pool P is drawn independently of them, the winner i of P satisfies

$$\Pr\left(a_i = 1 \mid \ell_i,\ s,\ i = \arg\max_{j \in P} \ell_j\right) = \pi(\ell_i, s).$$

However unequal the noise scales: heteroskedastic noise alone cannot bias committed confidence.

Proposition 2 · Exact split of the committed-token gap

$$\Delta(m) = \underbrace{\mathbb{E}_m[\sigma(\ell) - \pi(\ell)]}_{\text{own miscalibration}} + \underbrace{\mathbb{E}_m[\pi(\ell) - \pi(\ell, s)]}_{\text{field term}}$$

$\pi(\ell)$: ordinary calibration curve; $\pi(\ell, s)$: its context-aware version. At $m = 1$ winners are random, so the field term is exactly 0.

The field term is nonzero only when accuracy at fixed confidence depends on context: the **winner's curse in a corrected, testable form**.

What we will measure

E1 · committed-token gap

$\Delta(m) = \mathbb{E}_m[c_i - a_i]$ over tokens that win pools of size $m \in \{1, 2, 4, \ldots, 32\}$. Primary contrast $\Delta(32) - \Delta(1)$

E2 · field effect

Observed $\Delta(m)$ minus the gap predicted by $\hat{\pi}(\ell)$, which sees only the token's own confidence. A context-aware $\hat{\pi}(\ell, s)$ should cover the rest.

E3 · trajectory shift

Training uses uniform masks; decoders fill easy positions first. Are decoder states less calibrated at equal confidence and matched mask rate?

Hypotheses. **H1** $\Delta(32) - \Delta(1) > 0$; **H2** field term at $m = 32$ is positive; **H3** decoder states are more overconfident than uniform ones.

Diagnostics kept from the proposal, fitted on winning tokens:

$$\Pr(a_i{=}1) = \sigma\big(\alpha\,[+\alpha_s] + f(\ell_i) + \beta_z z_{r,m}\big),$$

or rank dummies γ_k in place of $z_{r,m}$. State effects α_s absorb the field term: $\beta_z < 0$ without them and ≈ 0 with them signals a field effect.

Three contributions on three models

C1
Selection-conditional calibration audit
On a 110M masked diffusion LM we train ourselves (so we control its training data), LLaDA-8B and Dream-7B.

C2
Both terms of the KL split
Exactly, in nats, on a synthetic JSON probe where p^* is known; as identified contrasts on real text.

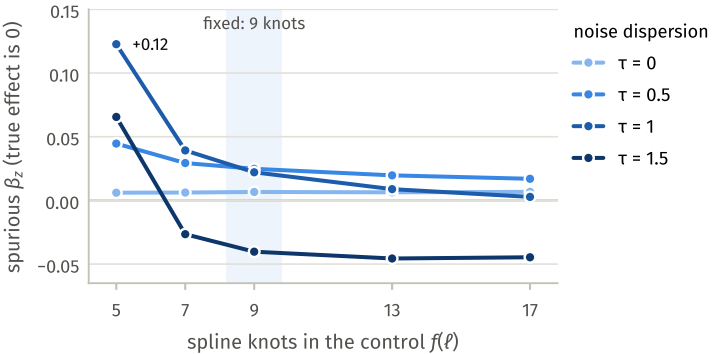
C3
A recalibration
Uses only quantities the decoder already computes: no extra forward passes.

EVALUATIONS SO FAR

Validate the instrument, then point it at a model

1 · estimator on known truth
2 · four simulated worlds (T0)
3 · determinism
4 · LLaDA-8B-Base pilot (T2)
5 · power
6 · our 110M model

Evaluation 1: the regression estimator on known truth

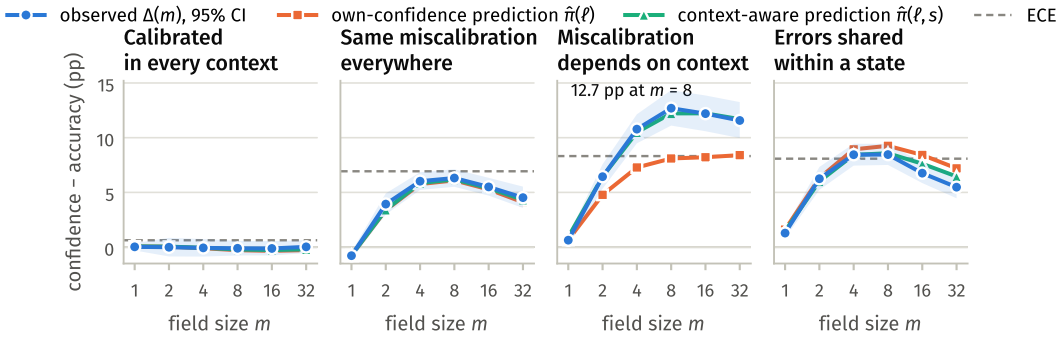


Heteroskedastic null: the noise scale varies across positions with dispersion τ, and there is no rank effect, so every nonzero β_z is spurious.

Planted β_z	Recovered
0 (null)	−0.017
−0.10	−0.061
−0.30	−0.242

- Planted effects come back smaller: **conservative**.
- 5 knots give a **spurious** +0.12 at τ = 1.
- **Decision:** 9 knots fixed in advance (5, 13 as sensitivity); β_z demoted to a diagnostic.

Evaluation 2 (T0): four simulations



Why ECE is not enough. With context-dependent miscalibration, ECE is 8.3 pp but a token that beat 8 candidates is **12.7 pp** overconfident; own confidence predicts 8.1, context recovers 12.2.

T0 metrics

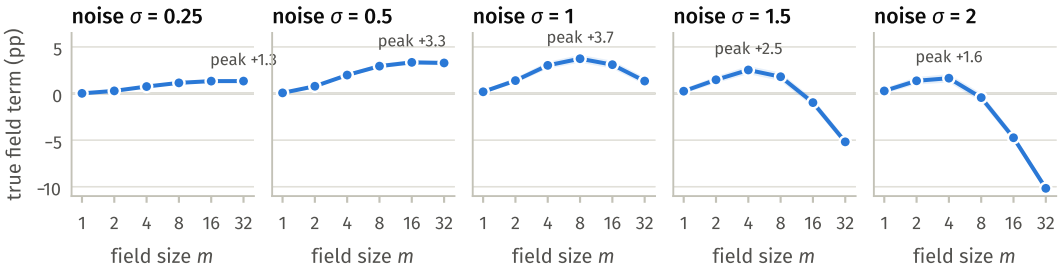
Δ and field term in pp, β_z in log-odds, with 95% intervals. Shaded: observed sign differs from the plan

Simulated world	Δ(32) − Δ(1)	Field term, m = 32	β_z, no FE	β_z, with FE
Calibrated in every context	0.0 [−0.7, +0.7] no / no	+0.1 [+0.1, +0.1] 0 / > 0	+0.010 [−0.055, +0.080] 0 / 0	+0.005 [−0.061, +0.076] 0 / 0
plan / observed				
Same miscalibration everywhere	+5.3 [+4.4, +6.3] yes / yes	+0.2 [+0.2, +0.3] 0 / > 0	−0.006 [−0.066, +0.056] 0 / 0	−0.004 [−0.066, +0.058] 0 / 0
plan / observed				
Miscalibration depends on context	+10.9 [+9.5, +12.4] yes / yes	+3.3 [+2.2, +4.4] > 0 / > 0	**−0.246** [−0.305, −0.184] < 0 / < 0	+0.062 [+0.001, +0.126] 0 / > 0
plan / observed				
Errors shared within a state	+4.2 [+3.2, +5.3] yes / yes	−0.7 [−0.9, −0.5] ? / < 0	+0.095 [+0.034, +0.156] < 0 / > 0	+0.030 [−0.033, +0.093] ≠ 0 / 0
plan / observed				

Δ(32) − Δ(1) is zero **only** under calibration, and β_z without fixed effects is strongly negative **only** when miscalibration depends on context, the true winner's curse. The shaded mismatches are on the next slide.

T0

True field term against field size at five noise levels (known truth, context-dependent world)



1. **The field-term estimator reads high**: +0.1 to +0.2 pp when the truth is zero, +3.3 pp when the truth is +1.6 pp[+0.4, +2.7].
2. **Fixed effects do not fully clean β_z** under context dependence: +0.062 [+0.001, +0.126].
3. **Shared errors give a negative field term with a positive β_z.** The true field term is not monotone in m: it peaks at m = 4–8 once noise is large and is negative at m = 32 for $\sigma \geq 1.5$.

Evaluation 3: the traces are deterministic

0

repeats are bitwise identical in ℓ, agreement and $\Delta(m)$

1,480

forward passes in the T2 pilot: 192,088 position rows in 195 s

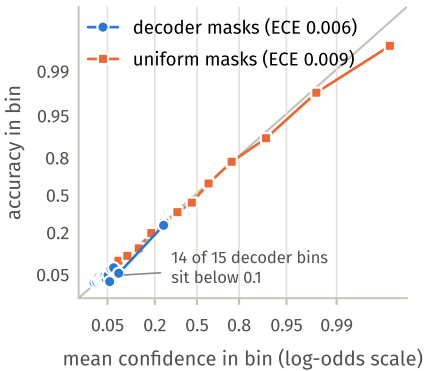
7.6/s

256-token forward passes, LLaDA-8B in fp16 split over 2 × RTX 2080 Ti

≈34 h

per 2,000-document arm under the full design (466 passes per document)

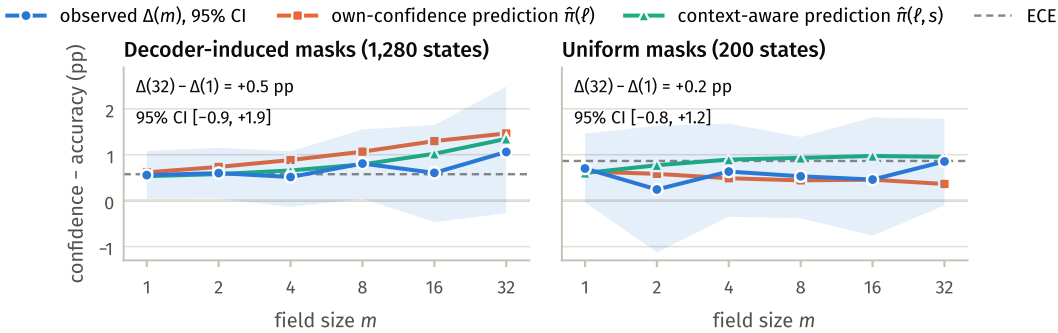
Evaluation 4 (T2 pilot): LLaDA-8B-Base is close to calibrated



pp unless noted	Decoder	Uniform
States	1,280	200
ECE	0.6	0.9
Δ(1), Δ(32)	0.6, 1.1	0.7, 0.9
Δ(32) − Δ(1)	+0.5 [−0.9, +1.9]	+0.2 [−0.8, +1.2]
Field(32) − field(1)	−0.04 [−0.14, +0.07]	+0.62 [+0.52, +0.72]
Residual ICC (ratio)	0.007	0.005
σ_d of the contrast	3.2	2.3

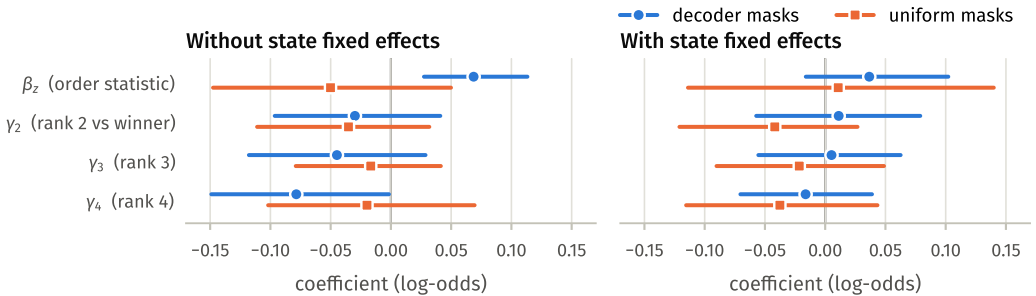
- Bins hug the diagonal in both arms. Decoder positions sit at low confidence because most decoder states are heavily masked.

T2: the committed-token gap does not grow with field size



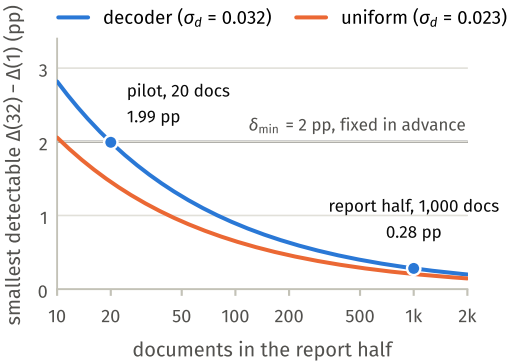
- Δ(*m*) stays within 0.2–1.1 pp at every *m*
- The uniform field contrast (+0.6 pp) excludes zero

T2 diagnostics: the decoder arm carries T0's shared-error signature



- Decoder arm: β_z without FE is **positive**, +0.069 [+0.027, +0.113], shrinking with FE
- At equal confidence, rank 4 is less accurate than the winner: γ_4 = −0.079 [−0.149, −0.002]. **Exploratory**: one of three γ_k, uncorrected.

Evaluation 5: the full sample can see effects of about 0.3 points



80% power, two-sided 5%: $n = 7.85\,\sigma_d^2/\delta^2$, with σ_d the between-document s.d. of the contrast in the pilot.

- A **1,000-document report** gives a standard error of about 0.1 pp on the contrast: effects near **0.3 pp** are detectable.

> **Gate**: If the standard error is of a detectable range, we will run the full plan

Evaluation 6: our 110M masked diffusion LM

Trained on FineWeb-Edu with our 2,000 calibration documents held out

Model	bidirectional transformer: 12 layers, width 768, 12 heads, learned positions
Tokens	GPT-2 BPE + [MASK]; sequence length 256
Objective	MDLM (weighted masked cross-entropy)
Optimiser	AdamW, lr 3×10^{-4}, 2,000 warm-up, decay to 10%, EMA 0.9999, fp16
v1 run	76,293 steps × 131,072 tokens = 10.0B tokens

38.4

perplexity *bound* (NELBO) on the 2,000 held-out docs [37.7, 39.2]

28.5

GPT-2 small (124M), *exact* perplexity, same docs [28.0, 29.0]

An expected gap: MDLM gives only an upper bound, and GPT-2 saw other data with far more compute. The fair reference is a same-size AR model on our data, trained by the same code.

Sample: locally fluent, globally incoherent. *"Maintain aeration while the protects food and bottom packaging wastes."*

Status by task

Task	Status	What exists	Remaining
Stage 0 (proposal era)	● done	LLaDA-8B 4-bit, 20 docs; both gates recorded	superseded by T2
T0 theory, simulations	◕ mostly	Propositions 1–2; four-world table and figure	second-person proof check; gate G0
T1 trace pipeline	● done	schema v2, three mask kinds, sharded and resumable; floor 0	—
T2 pilot, power	◕ mostly	LLaDA-8B-Base fp16 pilot; gate G1 = full plan	split manifest; freeze pre-registration
T3 110M model	◑ partly	v1 trained and evaluated; AR reference code	v2 retrain and evaluation, G2, AR run, release
T4 contamination	◑ partly	screens on both 8B base models	pre-register threshold; time-split set
T5–T7 E1, E2, E3	○ not started	estimators, calibrators, pilot numbers	full 2,000-document sweep
T8 controls	○ not started	precision adapters; PC-Sampler weight	precision, position, token type, temperature
T9–T10 C2	○ not started	fine-tuning hook; *p*(true) and top-20 in traces	probe generator, exact terms, real-text proxy
T11–T12 C3, decoders	◑ partly	decode loop, six commit policies; tasks formatted	recalibration rule; task runs

Summary

1 · The proposal was refined

Selection alone cannot bias a calibrated posterior (Proposition 1). Any bias lives in the field term (Proposition 2), which the instrument measures directly.

2 · The instrument works on known truth

The endpoints separate four simulated worlds where ECE cannot. Known weaknesses are fixed in advance (9 knots) or flagged (the field estimator reads high).

3 · First real model: near-calibrated, no gradient

LLaDA-8B-Base fp16: ECE 0.6 pp and $\Delta(32) - \Delta(1) = +0.5$ pp$[-0.9, +1.9]$. The decoder arm shows a shared-error signature.

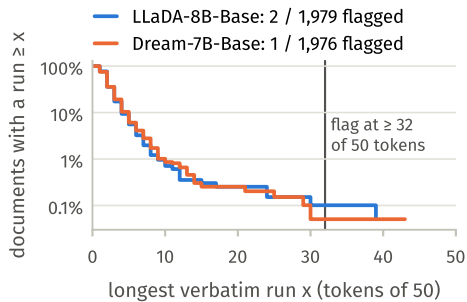
4 · Ready to scale

Deterministic traces, a written G1 gate, and a design that detects 0.3-point effects. Next: the big 2,000-document sweep on three models.

Backup: held-out text with ground truth at every position

- **2,000 calibration documents**, reservoir-sampled from FineWeb-Edu (fixed seed, ≥ 1,200 characters) and excluded from our 110M's training stream; **500 reserve documents** for the pilot.
- **50/50 document split**: calibrators, splines and C3 fitted on half 1; every reported number from half 2.
- Regenerated off-cluster: a **byte-identical** manifest.
- **Tasks**: GSM8K (1,319) and MBPP (500) as teacher-forced documents; HumanEval (164) for end-task evaluation only.

Contamination screen (T4)



Prompt with the first 50 tokens, generate 50 greedily, take the longest verbatim match. Median run: 1 token. The threshold of 32 must be pre-registered before flagged documents enter any analysis.

Backup: proofs

W_i: the event that $i \in P$ wins, $i = \arg\max_{j \in P} \ell_j$; ties broken by independent randomness U

Proposition 1

Suppose, given s, each pair (ℓ_i, a_i) is independent of $\ell_{-i} = (\ell_j)_{j \in M \setminus \{i\}}$. Let $Z = (\ell_i, s)$ and $Y = (\ell_{-i}, P, U)$.

Given s, $(\ell_i, a_i) \perp \ell_{-i}$ and $(P, U) \perp (\ell_j, a_j)_{j \in M}$, so $(\ell_i, a_i) \perp Y \mid s$, and by weak union $a_i \perp Y \mid Z$, i.e. $\mathbb{E}[a_i \mid Y, Z] = \pi(\ell_i, s)$.

W_i is determined by (Y, Z), so for every bounded measurable h

$$\mathbb{E}[a_i h(Z) 1_{W_i}] = \mathbb{E}[\pi(\ell_i, s) h(Z) 1_{W_i}].$$

Dividing by $\Pr(W_i) > 0$ gives $\mathbb{E}[a_i \mid Z, W_i] = \pi(\ell_i, s)$ a.s.

Proposition 2

Exactly one W_i holds. Applying the display with $h \equiv 1$, summing over i and averaging over s:

$$\mathbb{E}_m[a] = \sum_i \mathbb{E}_m[a_i 1_{W_i}] = \sum_i \mathbb{E}_m[\pi(\ell_i, s) 1_{W_i}] = \mathbb{E}_m[\pi(\ell, s)],$$

so $\Delta(m) = \mathbb{E}_m[\sigma(\ell) - \pi(\ell, s)]$; add and subtract $\pi(\ell)$.

At $m = 1$, $\mathbb{E}_1[\pi(\ell)] = \mathbb{E}_1[a]$ by the tower property and $\mathbb{E}_1[\pi(\ell, s)] = \mathbb{E}_1[a]$ by the display, so the field term is 0.

Backup: T2 pilot, full table

LLaDA-8B-Base fp16, 20 documents; probability units as in the report; 95% document-bootstrap intervals

	Decoder	Uniform
States	1,280	200
ECE	0.006	0.009
Δ(1)	0.006	0.007
Δ(32)	0.011	0.009
Δ(32) − Δ(1)	+0.005 [−0.009, 0.019]	+0.002 [−0.008, 0.012]
Field(32) − field(1)	−0.0004 [−0.0014, 0.0007]	+0.006 [0.005, 0.007]
β_z, no FE	+0.069 [0.027, 0.113]	−0.050 [−0.148, 0.050]
β_z, with FE	+0.037 [−0.016, 0.102]	+0.011 [−0.114, 0.140]
Residual ICC	0.007	0.005
σ_d of contrast	0.032	0.023
Documents for 80% power	20	11

Backup: what the traces cost

Schema v2, per 256-token document; bytes measured on LLaDA-8B rows

Design	Rows	Passes
Full plan v2	60,390	466
Without matched uniform	31,334	242
T2 pilot preset	9,601	74

Stored top-k	Bytes per row	Full plan
top-20, fp32 probabilities	186	83 GB
top-10, fp32 probabilities	117	53 GB

- Every masked position stores v_i, c_i, ℓ_i, a_i, the true token's probability, entropy, margin, the top-20 ids and probabilities, the change in ℓ_i since the previous step, and the commit rank.
- The top-k list is more than 80% of the bytes.
- Bytes were measured on a toy whose top-20 below the true token is noise: an upper bound for real models.

Backup: references

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